

Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

Aditya Kinjawadekar¹, Surender Kannaiyan¹, and Saugata Sinha¹

¹Visvesvaraya National Institute of Technology, Nagpur, India

E-mail: kinjawadekaradi112@gmail.com

Abstract

Low-resource regional dialects pose severe challenges for Natural Language Processing (NLP) systems due to significant morpho-syntactic variation, non-standard orthography, and extreme scarcity of annotated parallel corpora. In this paper, we address the task of dialect normalization—converting non-standard regional dialectal text into Standard Written Marathi—focusing on three major Indo-Aryan regional dialects of Maharashtra: Southern Konkan, Northern Konkan, and Varhadi. To overcome severe data scarcity, we introduce an automated synthetic parallel data expansion pipeline combining large language model (LLM) generation using Gemma-4 (12B) with a multi-tier verification engine (rule-based script filtering and LLM semantic auditing), effectively doubling the clean parallel training corpus into verified synthetically expanded pairs across agriculture and finance domains. We conduct a rigorous empirical evaluation of state-of-the-art multilingual sequence-to-sequence transformer architectures—google/mT5-small (300M) and IndicBART (244M)—under an 85/15 stratified test split with 5-fold cross-validation across 8 dataset configurations. Empirical results demonstrate that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the synthetically expanded multi-dialect dataset (+12.39 BLEU over IndicBART), with individual dialect performance reaching 81.00 BLEU and 91.21 chrF++ on Varhadi. We present detailed architectural comparisons, data scaling ablation studies, and qualitative subword morphological error analyses, establishing a benchmark for Indic dialect text normalization.

Keywords (at least 4-5): Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modeling, Synthetic Data Augmentation, mT5, IndicBART, Text Standardization.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] has made it possible to develop systems capable of understanding and generating natural language. Open-weight models are increasingly deployed to process text across various contextual domains, as they allow finer control over model architecture, runtime environment, and training data [2]. Even with these advancements, current language systems predominantly cater to standard languages, yielding substandard outputs when queried with non-standard regional dialects [3]. Speakers of regional dialects routinely encounter severe performance degradation across automated speech recognition, machine translation, and text processing systems due to distinct lexical, phonetic, and morpho-syntactic variations [4–6].

Dialect normalization is the task of transducing non-standard regional dialectal text into standard written language while preserving original sentence semantics [5]. It represents an essential layer for digital linguistic inclusion. In linguistically diverse nations such as India—home to 22 officially recognized languages and hundreds of regional dialects spoken by 1.4 billion people—sub-regional varieties remain severely under-represented in computational research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [7]. Because its regional dialects possess distinct lex-

ical, phonetic, and morpho-syntactic properties, we hypothesize that model performance and evaluation metrics will vary significantly across different dialectal varieties.

Building pipelines for translating Indic dialects primarily comes down to two core challenges:

1. **Extreme Data Scarcity:** Datasets that map regional dialects to standard forms are technically non-existent; and
2. **Morphological Complexity:** Marathi is a language with rich morphology, where noun cases, postpositional affixes, and verbal tense suffixes exhibit complex variations [7, 8].

Addressing these limitations, this work presents a multi-dialect Marathi text normalization framework combining LLM-driven synthetic data augmentation with sequence-to-sequence transformer fine-tuning. We undertake this study because the performance of existing multilingual models on regional Marathi dialects remains largely unevaluated, and because automated synthetic data expansion—when constrained by multi-tier verification—can reliably enhance output accuracy for severely resource-constrained tasks.

Our core contributions are:

- **Parallel Corpus Generation:** Based on transcripts from the RESPIN dataset [9], we construct an aligned parallel corpus for three major Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi across agricultural and banking domains.
- **Automated Synthetic Augmentation with Verification:** We design a synthetic data expansion pipeline using Gemma-4 [10] paired with a multi-tier verification engine (rule-based script filtering and LLM semantic auditing), demonstrating that audited synthetic data significantly improves downstream model accuracy.
- **Cross-Dialect Evaluation & Model Benchmarking:** We empirically demonstrate that normalization performance varies substantially across dialects, and establish comprehensive benchmarks for existing multilingual models (mT5-small vs. IndicBART) across single-dialect and multi-dialect training configurations.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early work in European dialect normalization relied on Character-Level Statistical Machine Translation (CSMT) and phrase-based SMT for Swiss German Dialects [11], which were subsequently superseded by pre-trained language models and neural sequence-to-sequence architectures [12, 13]. Parallel efforts have addressed dialectal normalization across Uralic and Scandinavian varieties, including Finnish, Swedish, and Estonian [14–16]. A systematic multilingual evaluation across four language families by Kuparinen et al. [17] demonstrated that fine-tuned sequence-to-sequence transformers consistently outperform rule-based and statistical baselines. Beyond European languages, East Asian and Afro-Asiatic dialect processing has evolved along similar lines: Japanese regional dialects have been modeled via phrase-level LSTMs [18], Mandarin non-standard varieties via acoustic-phonetic mapping [19], and Dialectal Arabic via the Conventional Orthography for Dialectal Arabic (CODA) framework [20], enabling multi-city translation benchmarks (MADAR [21], NADI [22]) and morphology-aware prompting strategies [23, 24].

2.2 Indic Dialect Landscape and Marathi NLP

While pre-trained multilingual language models such as IndicBART [25], IndicNLP Suite [26], and mT5 [27] cover major Indic national languages, they suffer from significant performance degradation on unscripted regional sub-dialects. Monolingual suites like L3Cube-MahaNLP [28] provide pre-trained Marathi models for classification and tagging, but lack sequence transduction capabilities for generative text standardization. In the broader Indic context, INDIC-DIALECT [29] introduced translation benchmarks for Hindi variants (Bhojpuri, Magahi, Maithili)

evaluated using Weighted Finite-State Transducers (WFST) [30], while Dhasmana et al. [31] explored cross-lingual transfer for low-resource Indic varieties. However, research on Marathi regional dialects has remained largely restricted to speech/ASR classification [32, 33], zero-shot retrieval heuristics [34, 35], or preliminary rule-based text mapping [36]. Our work addresses this gap by presenting an end-to-end neural fine-tuning pipeline supported by multi-tier synthetic data augmentation and rigorous cross-dialect benchmarks.

2.3 LLM Distillation and Multi-Tier Verification

Large Language Models (LLMs) are increasingly leveraged to alleviate data scarcity across generation tasks [37–39]. Knowledge distillation and synthetic data expansion enable the training of compact, efficient task-specific models using outputs generated by larger teacher architectures [40–42]. However, applying unconstrained synthetic generation to low-resource regional dialects introduces substantial risks of hallucination, non-standard orthographic noise, and language leakage [43]. We mitigate these risks by combining Gemma-4 generation with a morphologically tailored prompt design and a closed-loop, multi-tier verification engine to guarantee dataset integrity prior to neural fine-tuning.

3 Methodology

3.1 Target Dialects and Linguistic Properties

The following Marathi dialects from the RESPIN corpus were selected for this study, as they exhibit unique lexical and phonetic variations:

1. **Southern Konkani:** Shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms are its key characteristics. Lexical differences include vowel shifts: standard जातो (“he goes”) becomes जातां; standard कुठे (“where”) becomes कुठंय.
2. **Northern Konkani:** Since it is surrounded by neighboring Gujarati and Bhili language families, it tends to exhibit significant consonant variations. Few examples are: standard कशाला (“why”) → Northern Konkani कसाले; standard मुलगा (“boy”) → Northern Konkani पोर.
3. **Varhadi:** Spoken in the regions of Vidarbha accompanied with verb suffix modifications. Standard questioning का? (“why?”) becomes काऊन?; verbal forms exhibit noticeable patterns. Varhadi maintains closest structural similarity to Standard Marathi as compared to other dialects since the transformations are more systematic.

3.2 Original Corpus Curation and Foundation in RESPIN

Our primary data source is the RESPIN-S1.0 initiative, developed by researchers at IISc Bangalore. RESPIN provides audio recordings and their transcripts in agriculture and banking domains for different Indian languages and their dialects. However, the corpus contains only raw dialect transcripts—it does not provide aligned dialect-to-standard Marathi translations, which are essential for training normalization models.

To bridge this gap, we used Gemma-4 (12B configuration, accessed via Ollama, run locally) to translate each dialect transcript into Standard Marathi, thereby constructing the original parallel corpus. The unique prompts from each dialect partition (5,838 for Southern Konkani, 5,825 for Northern Konkani, and 5,330 for Varhadi) were individually processed through the LLM translation pipeline described in Section 3.3, yielding 16,163 verified parallel pairs across three dialects.

3.3 LLM-Driven Translation and Verification Pipeline

Both the original parallel corpus construction (Phase 1) and the subsequent synthetic expansion (Phase 2) rely on the same LLM-driven translation and verification pipeline. In Phase 1, each RESPIN dialect transcript is translated into Standard Marathi to create the initial parallel pairs. In Phase 2, additional synthetic dialect–standard pairs are generated from the same source transcripts to expand the training corpus. The pipeline employs the following design

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkani (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkani (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	All Dialects	809,934 (100%)	1,026.06	2,644	23,069

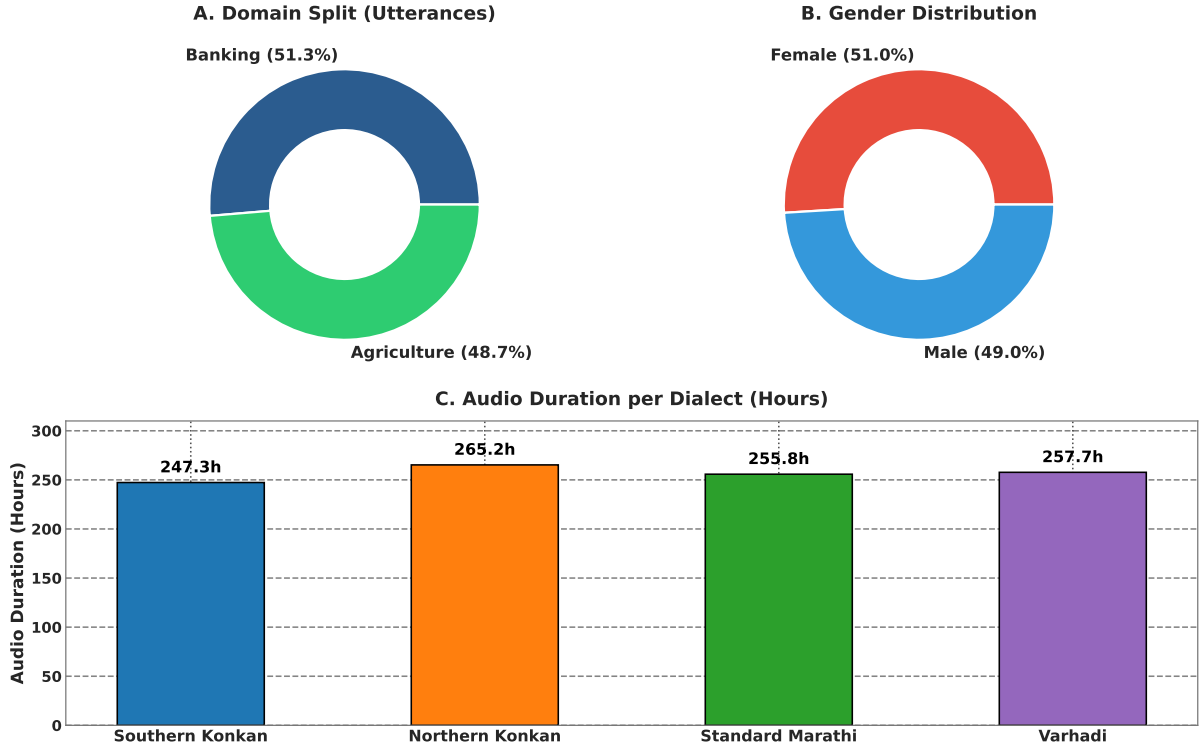


Figure 1: RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

aspects:

- **1-Prompt-Per-Sentence Architecture:** Instead of batching multiple pairs per API call (which risks contextual loss), each sentence is processed separately.
- **Detailed Prompting Strategy:** System prompt is designed for the output to preserve exact meaning, context regarding the domain, proper nouns, and numerical values from the input with a set of 5-8 few-shot examples per dialect.
- **Temperature Control:** Generation temperature is set to 0.3 with top- $p = 0.9$ for optimal results.

All translations are generated using Gemma-4 under the 12B configuration accessed via Ollama, run locally. Every generated pair—whether from Phase 1 or Phase 2—undergoes a three-tier verification and correction pipeline:

1. Tier 1: Rule-Based Syntactic Filter.

- **Devanagari Script Integrity:** Both source and target must have at least 90% Devanagari Unicode characters that lie in the U+0900--U+097F range, that automatically get rid of unsolicited characters.
- **Length Ratio Constraint:** $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
- **Non-Identity Constraint:** $L_{\text{dialect}} \neq L_{\text{standard}}$ to flag identical string copies.
- **Minimum Token Count:** Both source and target must contain at least 3 whitespace-separated tokens.

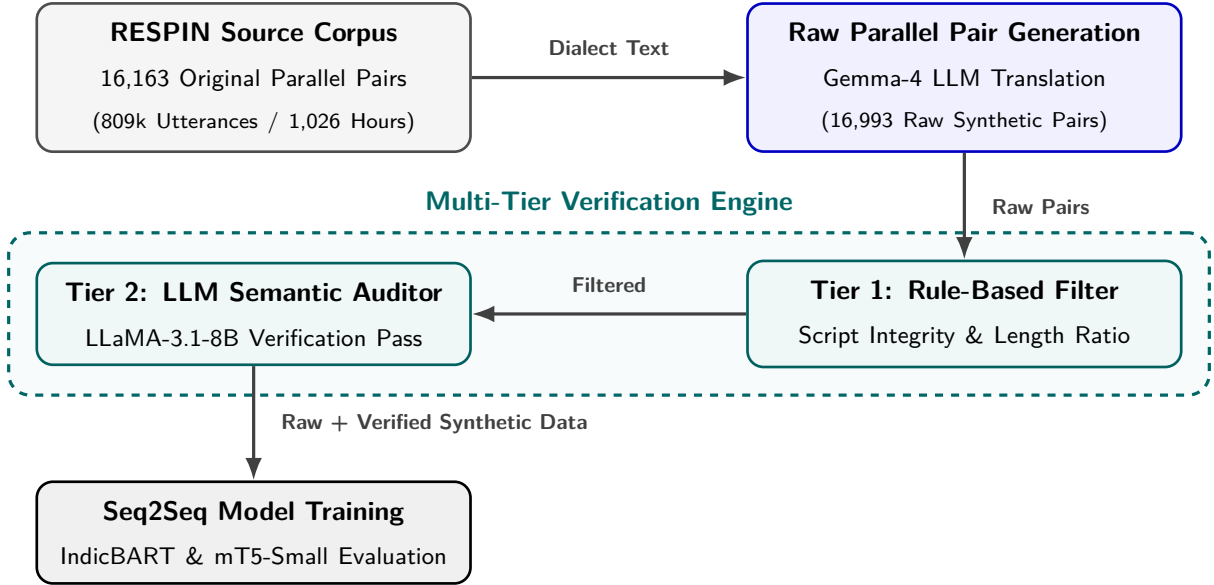


Figure 2: System architecture of the multi-dialect normalization framework.

Table 2: Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkani	5,576	5,838	5,569	269 (4.61%)	11,145
Northern Konkani	5,501	5,825	5,534	291 (5.00%)	11,035
Varhadi	5,086	5,330	5,069	261 (4.90%)	10,155
Total	16,163	16,993	16,172	821 (4.83%)	32,335

- Tier 2: LLM Semantic Verification.** A secondary round of evaluation was performed using llama-3.1-8b-instant (accessed via Groq API). The verification happens on a scale of 1–5 score that considers semantics, hallucinations, multilingual contamination, and complete standardization of dialects. Pairs that score < 4.0 are rejected or routed for further correction.
- Closed-Loop Correction Engine.** Pairs that are flagged by Tier 1 or Tier 2 enter a correction loop: Step 1 generates corrected translations using specialised prompts based on the pre-defined reason of identified failure (e.g., “Hindi leakage detected” or “Incomplete normalization”); Step 2 verifies the corrected sentences again, following the complete discard of pairs that fail the second audit pass.

3.4 Synthetic Data Expansion (Phase 2)

After constructing the original 16,163 verified parallel pairs through Phase 1, we applied the same pipeline to synthetically expand the training corpus. The expansion phase re-processes the source dialect transcripts under varied prompting conditions to generate additional dialect-standard pairs, effectively doubling the training data.

The pipeline was used to generate parallel pairs from the RESPIN transcripts and for synthetic data expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

3.5 Evaluation Protocol

For stable performance during development, models were validated using 5-Fold Cross-Validation on an 85/15 stratified split of the train set. The final reports were generated on the official held-out test set provided by RESPIN after training the model on the complete dataset,

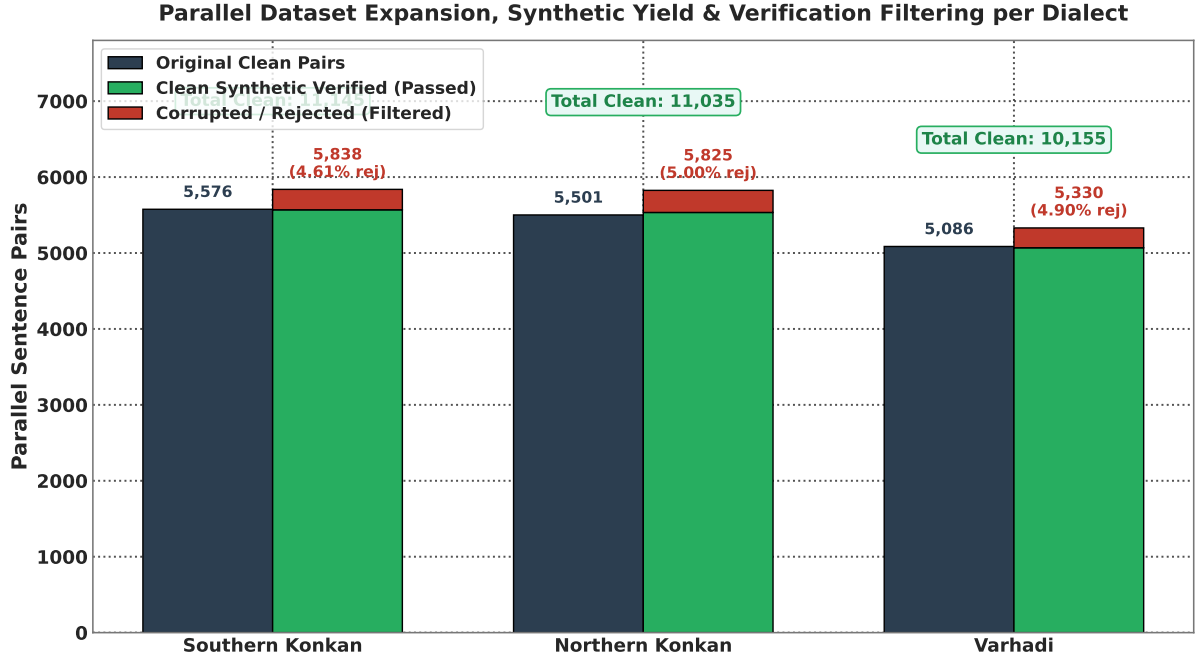


Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

which had around 2,170 utterances across target dialects. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions. Results are reported using metrics that are evaluated using SacreBLEU [44] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.
- **Normalized WER / CER**: Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

4.1 Comprehensive Test Benchmark on IISc_RESPIN_test_mr

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance**: mT5-small upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling**: Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.

4.2 Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

Table 3: Comprehensive Evaluation Matrix on IISc_RESPIN_test_mr Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkani	559	57.80	26.60%	43.86	34.37%
		Northern Konkani	540	90.13	6.46%	79.46	11.95%
		Varhadi	516	83.59	10.19%	74.81	14.73%
	Synthetically Expanded	Southern Konkani	559	24.06	60.32%	44.36	32.75%
		Northern Konkani	540	65.41	28.70%	79.76	12.61%
		Varhadi	516	67.97	25.43%	76.90	14.19%
Multi-Dialect	Original	Southern Konkani	559	26.21	53.15%	40.94	35.34%
		Northern Konkani	540	47.59	32.84%	77.50	14.65%
		Standard Benchmark	555	96.12	3.12%	96.65	1.91%
		Varhadi	516	72.75	25.30%	73.47	16.16%
	Synthetically Expanded	Southern Konkani	559	52.06	36.80%	43.88	34.96%
		Northern Konkani	540	49.08	28.70%	78.16	13.55%
		Standard Benchmark	555	96.80	1.65%	97.39	1.37%
		Varhadi	516	70.27	21.40%	74.74	15.57%

Table 4: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	5,576	837 / 836	48.54	78.35	46.31	74.43
Northern Konkani (D2)	5,501	826 / 825	60.58	80.30	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	81.00	91.21

4.3 Impact of Synthetic Data Augmentation

Table 8 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming that data scarcity was a primary performance bottleneck.

But a dialect-wise breakdown suggests notable variance across the corpora. Southern Konkani showed significant gains from expansion (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small), while Northern Konkani sees moderate gains (+1.49 BLEU for IndicBART, +0.37 BLEU for mT5-small) due to its already high original baseline performance. We attribute this behavior to two factors:

1. **Baseline Saturation and Precision Dilution:** Varhadi already had better performance on original non-expanded set (72.75-79.57 BLEU). We estimate that LLM-generated synthetic pairs may introduce suffix rules which were over-generalized that slightly degraded the exact N-gram matching against the original reference texts.
2. **Cross-Dialect Parameter Competition:** In shared dialect training, we estimate that appending synthetic data across distinct sub-dialects may have caused performance competition in shared attention weights, which could have considered prioritizing recovery on severely sparse dialects (Southern Konkani) over saturated ones.

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,742 rejected pairs that contained garbage translations. Table 9 presents the

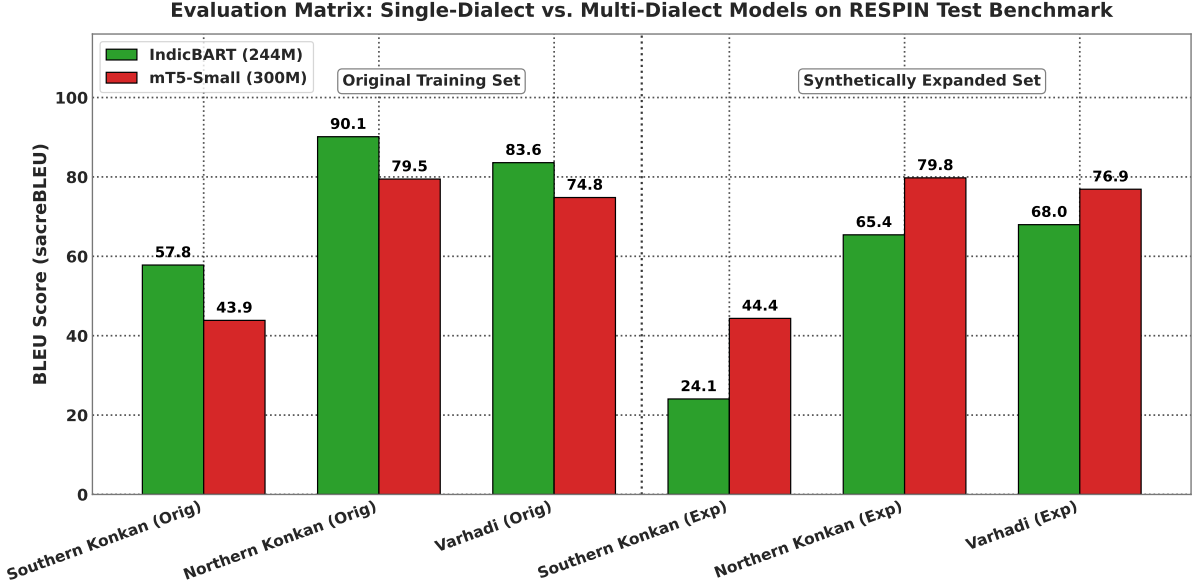


Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

Table 5: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkan (D1)	16,163	837 / 836	26.21	53.15	48.28	75.04
Northern Konkan (D2)	16,163	826 / 798	47.59	67.16	62.35	78.92
Varhadi (D4)	16,163	763 / 790	72.75	85.63	79.57	90.51
Overall Combined	16,163	2,426 / 2,424	48.17	68.58	63.29	81.37

quantitative comparison on the benchmark suite.

Key findings and insights from this evaluation include:

1. **Quality Collapse on Unverified Data:** Training IndicBART on unverified outputs causes a significant drop of 14.03 BLEU points, collapsing from 57.12 to 43.09 due to compounded error propagation from uncorrected dialect residue and wrong syntax with other errors.
2. **mT5-Small Resilience:** While google/mT5-small showed resilience due to its 250,000-token multilingual capacity, training on unverified data still suffers an 8.30 BLEU point drop, falling from 69.51 to 61.21 and a 4.40 chrF++ drop.
3. **Essential Role of Multi-Tier Auditing:** Removal of flawed ones and correcting the possible set of pairs is directly responsible for better performance that eliminated subsequent errors.

4.5 Model Architecture Analysis

Table 10 provides a detailed architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage:** mT5’s 250K vocabulary provides $3.9\times$ more subword units than IndicBART’s 64K, enabling finer-grained tokenization of morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix

Table 6: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	11,145	1,672 / 1,671	47.25	64.69	65.10	82.88
Northern Konkani (D2)	11,035	1,656 / 1,655	40.51	62.21	62.07	79.29
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	78.89	90.59

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialect-wise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	32,335	1,672 / 1,710	52.06	72.15	66.62	83.57
Northern Konkani (D2)	32,335	1,656 / 1,627	49.08	69.90	62.72	79.63
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	78.73	90.46
Overall Combined	32,335	4,852 / 4,850	57.12	75.49	69.51	84.58

changes must be captured at the subword level.

2. **Positional Encoding:** mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead:** IndicBART requires explicit language tokens ($\langle 2_{mr} \rangle$) that add conditioning complexity, whereas mT5’s pure text-to-text format enables direct mapping without task-specific architectural constraints.

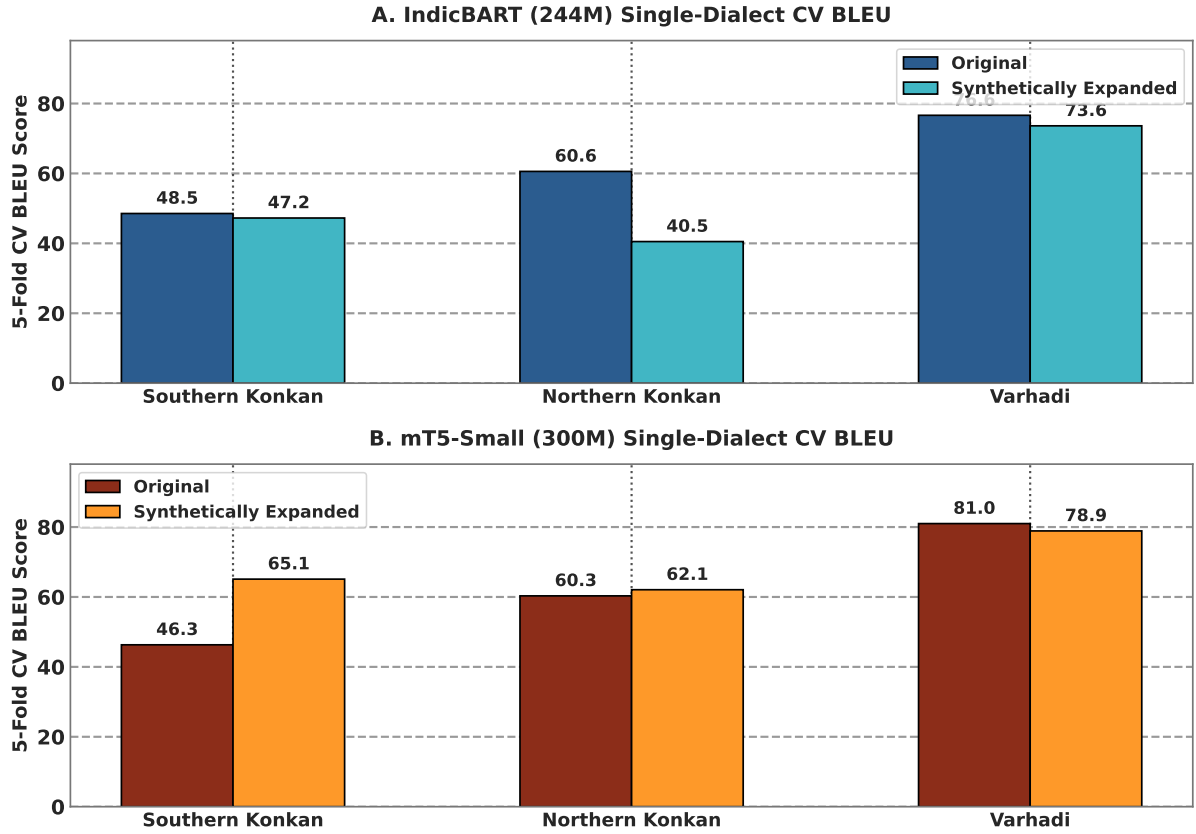


Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

Table 8: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkani (D1)	26.21	52.06	+25.85	48.28	66.62	+18.34
Northern Konkani (D2)	47.59	49.08	+1.49	62.35	62.72	+0.37
Varhadi (D4)	72.75	70.27	-2.48	79.57	78.73	-0.84
Overall Combined	48.17	57.12	+8.95	63.29	69.51	+6.22

Table 9: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	-14.03	-10.95
	Filtered Clean Data	32,335	57.12	75.49	+14.03	+10.95
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	-8.30	-4.40
	Filtered Clean Data	32,335	69.51	84.58	+8.30	+4.40

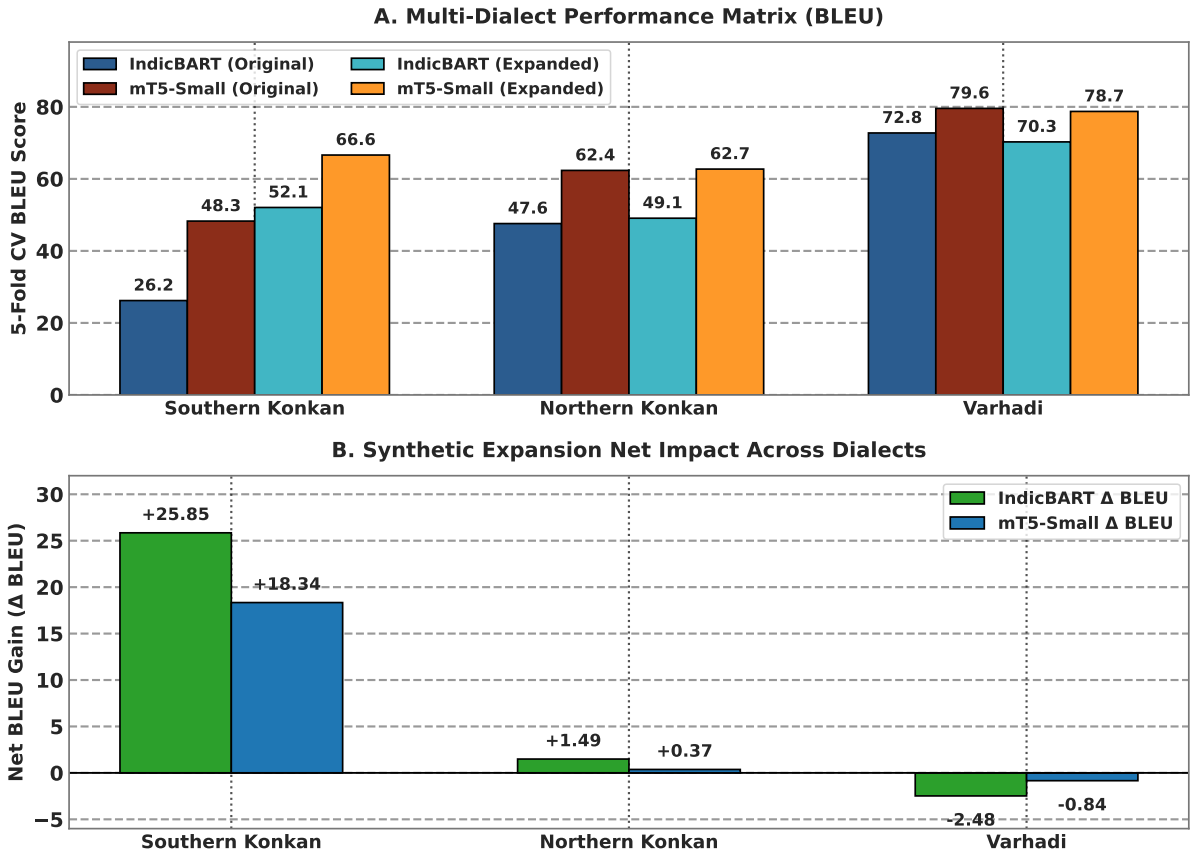


Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

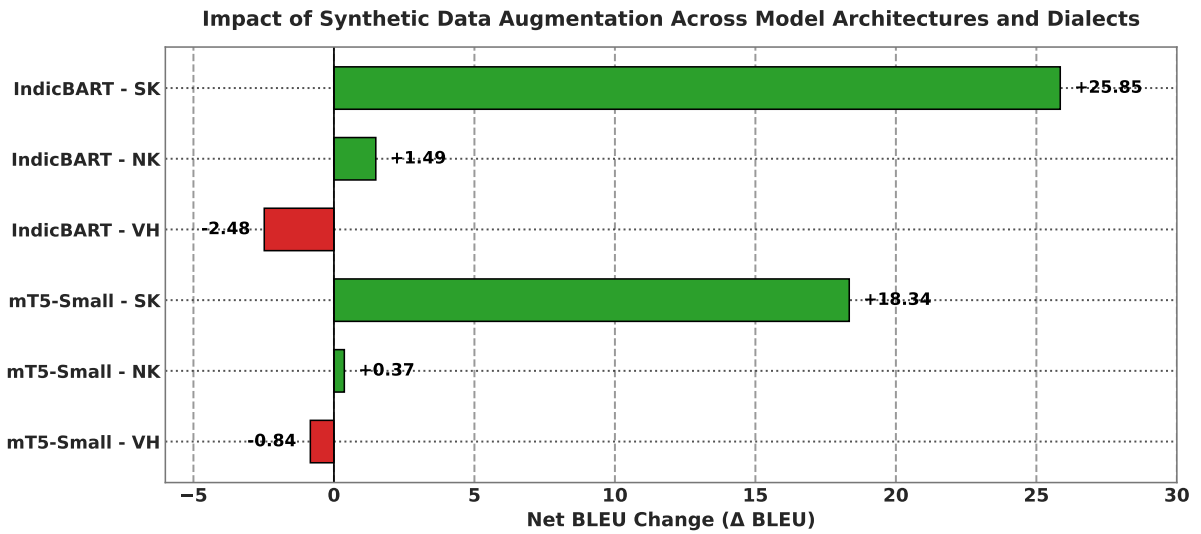


Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

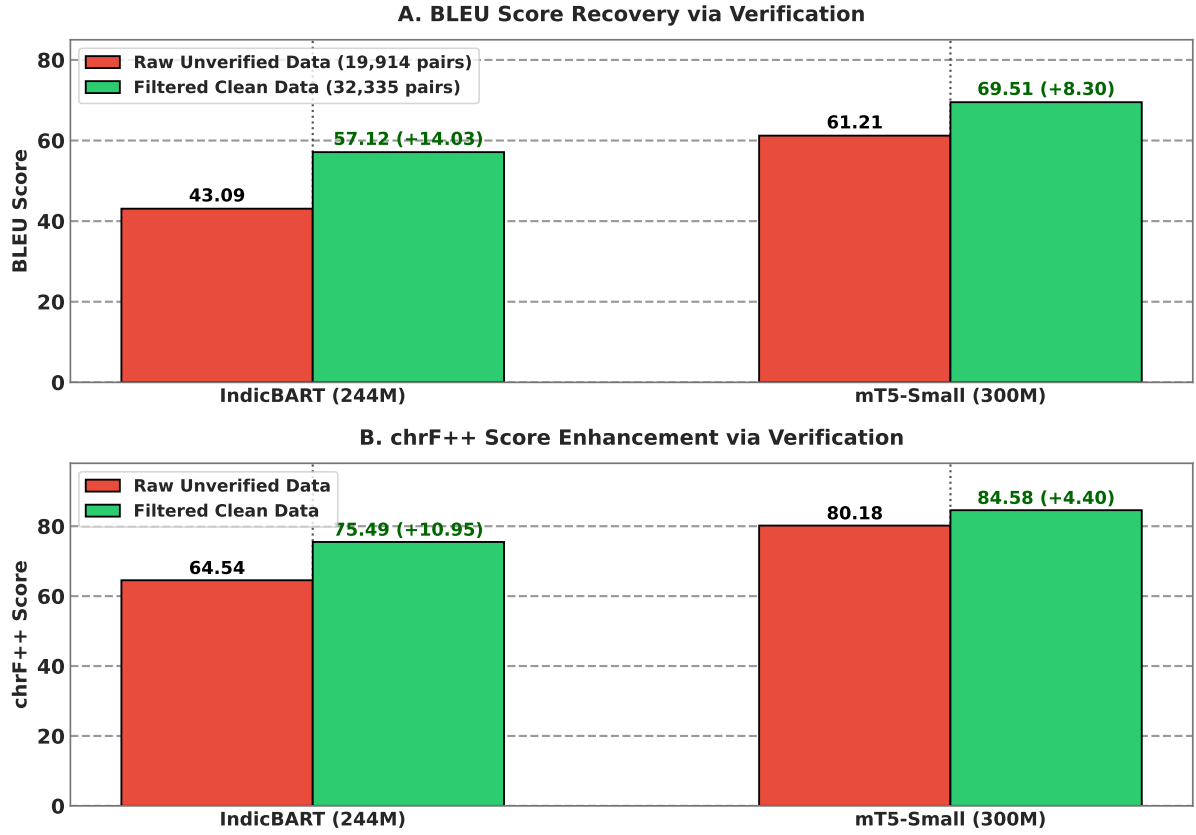


Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Table 10: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required $\langle 2_{mr} \rangle$ token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

5 Conclusion

This study presents a comprehensive, reproducible framework for multi-dialect Marathi text normalization, investigating dialectal variance under normalization, the efficacy of verified synthetic augmentation, and the performance of pre-trained multilingual sequence-to-sequence models on low-resource Indic dialects.

Our empirical findings confirm that normalization difficulty varies significantly across regional dialects: individual dialect performance ranged from 66.62 BLEU (Southern Konkani) to 78.73 BLEU (Varhadi) for mT5-small on the expanded multi-dialect benchmark, highlighting the necessity of dialect-specific evaluation rather than treating Marathi as a monolithic target.

Our LLM-driven synthetic data augmentation pipeline, powered by Gemma-4 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing), successfully expanded the clean parallel corpus from 16,163 to 32,335 pairs while maintaining high semantic fidelity. Multi-tier verification proved crucial for augmentation quality, yielding a recovery of +14.03 BLEU for IndicBART and +8.30 BLEU for mT5-small compared to training on raw, unverified synthetic outputs.

Evaluating existing pre-trained multilingual architectures on the expanded benchmark established new state-of-the-art baselines: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. This performance margin is largely driven by mT5’s $3.9\times$ larger subword vocabulary (250K vs. 64K tokens) and relative positional biases, which accommodate the morphological complexity of dialectal Marathi more effectively.

5.1 Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on LLM inference pipelines (Gemma-4 via local/hosted endpoints), introducing computational resource and latency considerations for large-scale replication.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-4) have not been evaluated for direct zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on intrinsic text normalization metrics (BLEU, chrF++, WER); downstream task evaluations (e.g., ASR post-processing, translation) remain for future work.
4. Synthetic augmentation exhibited a slight negative yield on Varhadi (-0.84 BLEU for mT5-small), suggesting that for already high-performing dialects, over-generalized suffix rules in synthetic data may slightly dilute precision—highlighting the need for targeted augmentation strategies.

References

- [1] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, pages 5998–6008, 2017. URL <https://arxiv.org/abs/1706.03762>.
- [2] Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, and et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL <https://arxiv.org/abs/2407.21783>.
- [3] Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 6282–6293, 2020. doi: 10.18653/v1/2020.acl-main.560.
- [4] Allison Koenecke, Andrew Nam, Emily Lake, Joe Nudell, Minnie Quartey, Zion Mengesha, Connor Touns, John R Rickford, Dan Jurafsky, and Sharad Goel. Racial disparities in automated speech recognition. *Proceedings of the National Academy of Sciences*, 117(14):7684–7689, 2020.

- [5] Verena Blaschke, Christoph Purschke, Hinrich Schütze, and Barbara Plank. What do dialect speakers want? a survey of attitudes towards language technology for german dialects. *arXiv preprint arXiv:2402.11968*, 2024. URL <https://arxiv.org/abs/2402.11968>.
- [6] Aditya Joshi, Raj Dabre, Diptesh Kanojia, Zhuang Li, Haolan Zhan, Gholamreza Haffari, and Doris Dippold. Natural language processing for dialects of a language: A survey. *ACM Computing Surveys*, 57(6):1–37, 2025. URL <https://doi.org/10.1145/3712060>.
- [7] Sonal Kulkarni-Joshi. Variation and change in dialects of marathi: A social-dialectological approach. In Pritha Chandra, editor, *Variation in South Asian Languages: From Macro to Micro-Differences*, pages 207–236. Springer, 2023. doi: 10.1007/978-981-99-1149-3_9.
- [8] Nazmuddoha Ansary, Quazi Adibur Rahman Adib, Tahsin Reasat, Asif Shahriyar Sushmit, Ahmed Imtiaz Humayun, Sazia Mehnaz, Kanij Fatema, Mohammad Mamun Or Rashid, and Farig Sadeque. Unicode normalization and grapheme parsing of indic languages. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 17019–17030. ELRA Language Resource Association, 2024.
- [9] Saurabh Kumar, Abhayjeet Singh, Deekshitha G, Amartyaveer, Jesuraja Bandekar, Savitha Murthy, Sumit Sharma, Sandhya Badiger, Sathvik Udupa, Amala Nagireddi, Srinivasa K. M. Raghavan, Rohan Saxena, Jai Nanavati, Raoul Nanavati, Janani Sridharan, Arjun Mehta, Ashish K. S. Khuraishi, Sai Praneeth Reddy Mora, Prashanthi Venkataramakrishnan, Gauri Date, Karthika P, and Prasanta Kumar Ghosh. RESPIN-S1.0: A read speech corpus of 10000+ hours in dialects of nine indian languages. In *Advances in Neural Information Processing Systems (NeurIPS 2025) Track on Datasets and Benchmarks*, pages 1–13, 2025.
- [10] Google DeepMind Gemma Team. Gemma 4: Open language models. *arXiv preprint arXiv:2408.00118*, 2024. URL <https://arxiv.org/abs/2408.00118>.
- [11] Yves Scherrer and Nikola Ljubešić. Normalising swiss german dialect text using character-level machine translation. In *Proceedings of the 3rd Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2016)*, pages 117–126, 2016.
- [12] Mathilde Lusetti, Tsetsu Ruzsics, Tanja Samardžić, and Yves Scherrer. Impact of target language models on swiss german dialect normalization. In *Proceedings of the 5th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2018)*, pages 133–141, 2018.
- [13] Mihael Kresic and Noorhan Abbas. Normalizing swiss german dialects with the power of large language models. *Procedia Computer Science*, 244:287–295, 2024.
- [14] Niko Partanen, Mika Härmäläinen, and Khalid Alnajjar. Dialect text normalization to normative standard finnish. In *Proceedings of the 5th Workshop on Noisy User-generated Text (W-NUT @ EMNLP 2019)*, pages 141–146. Association for Computational Linguistics, 2019.
- [15] Mika Härmäläinen, Niko Partanen, and Khalid Alnajjar. Normalizing swedish dialects spoken in finland. In *Proceedings of the 7th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2020)*, pages 55–62, 2020.
- [16] Mika Härmäläinen, Niko Partanen, and Jack Rueter. Generating dialectal finnish for dialect normalization of estonian dialects. In *Proceedings of the 9th Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ COLING 2022)*, pages 71–79, 2022.
- [17] Olli Kuparinen, Aleksandra Milić, and Yves Scherrer. Dialect-to-standard normalization: A large-scale multilingual evaluation. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 13814–13828. Association for Computational Linguistics, 2023.
- [18] Kaori Abe, Kentaro Inui, and Yuji Matsumoto. Multi-dialect neural machine translation for japanese dialects. In *Proceedings of the 27th International Conference on Computational Linguistics (COLING 2018)*, pages 2105–2116, 2018.
- [19] Yuan Zhao and Eleanor Chodroff. Acoustic-phonetic variation in mandarin dialect corpora. *Journal of Phonetics*, 94:101172, 2022.
- [20] Nizar Habash, Mona Diab, and Owen Rambow. Conventional orthography for dialectal arabic. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC '12)*, pages 711–718, 2012.
- [21] Houda Bouamor, Nizar Habash, Mohammad Salameh, et al. The madar parallel corpus of arabic dialects. In *Proceedings of the 11th Language Resources and Evaluation Conference (LREC 2018)*, pages 3375–3381, 2018.
- [22] Muhammad Abdul-Mageed, Chunsheng Zhang, AbdelRahim Elmadany, and Ebrahim Mohamed Nagoudi. Nadi 2023: The fourth nuanced arabic dialect identification and translation shared task. In *Proceedings of the Arabic NLP Conference (ArabicNLP @ EMNLP 2023)*, pages 512–528, 2023.
- [23] Fadhil Eryani, Nizar Habash, Faisal Al-Shargi, et al. A normalization corpus for five arabic dialects. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC 2020)*, pages 5750–5757, 2020.

- [24] Antonios Dimakis, John Pavlopoulos, and Antonios Anastasopoulos. Dialect normalization using large language models and morphological rules. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 23696–23714. Association for Computational Linguistics, 2025.
- [25] Raj Dabre, Himani Shrottriya, Anoop Kunchukuttan, Ratish Puduppully, Mitesh M. Khapra, and Pratyush Kumar. Indicbart: A pre-trained model for indic natural language generation. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 1849–1863, 2022. URL <https://doi.org/10.18653/v1/2022.findings-acl.145>.
- [26] Divyanshu Kakwani, Anoop Kunchukuttan, Satish Golla, N. C. Gokul, Pushpak Bhattacharyya, Mitesh M. Khapra, and Pratyush Kumar. Indicnlp suite: Monolingual corpora, evaluation benchmarks and pre-trained multilingual language models for indic languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4948–4961. Association for Computational Linguistics, 2020.
- [27] Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mt5: A massively multilingual pre-trained text-to-text transformer. In *Proceedings of NAACL-HLT 2021*, pages 483–498, 2021. URL <https://doi.org/10.18653/v1/2021.naacl-main.41>.
- [28] Raviraj Joshi, Parth Goel, and Rucha Joshi. L3cube-mahanlp: Marathi natural language processing datasets, models, and library. In *Proceedings of the WILDRE-6 Workshop, LREC 2022*, pages 10–17, 2022. URL <https://aclanthology.org/2022.wildre-1.2>.
- [29] Aman Kumar, Manish Shrivastava, and Pushpak Bhattacharyya. Indic-dialect: Benchmark dataset for indic dialect classification and translation. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 15201–15212. ELRA Language Resource Association, 2024.
- [30] Vineel Pratap, Ankit Saraf, and Pratyush Kumar. Wfst-guided neural text normalization for indic languages. In *Proceedings of Interspeech 2024*, pages 3120–3124, 2024.
- [31] Akriti Dhasmana, Aarohi Srivastava, and David Chiang. Dialect matters: Cross-lingual asr transfer for low-resource indic language varieties. In *Proceedings of the Thirteenth Workshop on NLP for Similar Languages, Varieties and Dialects (VarDial @ ACL 2026)*, pages 145–156. Association for Computational Linguistics, 2026.
- [32] Rahesha Mulla and B. Suresh Kumar. Text-independent automatic dialect recognition of marathi language using spectro-temporal characteristics of voice. *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 10(2s):313–321, 2022. doi: 10.17762/ijritcc.v10i2s.5949.
- [33] Poonam S. Pawar, Jatinderkumar R. Saini, and Shraddha Vaidya. Automatic marathi dialect classification in noisy and non-noisy environments. *Procedia Computer Science*, 282:288–303, 2026.
- [34] Gaurav Gaikwad, Deepak Panchal, Omprasad Deshmukh, and Sonali Patil. Marathi dialect to pure marathi using rag & ai. *International Research Journal of Engineering and Technology (IRJET)*, 12(11):787–791, 2025. ISSN 2395-0056.
- [35] V. A. Vyavhare, S. N. Divekar, Harshada Pawar, Karan Kale, and Subhash Jagtap. Marathi dialect translator. *International Journal of Scientific Research in Science, Engineering and Technology (IJSRSET)*, 13(3):254–261, 2026. doi: 10.32628/IJSRSET2613331.
- [36] Prasad Sanjay Gavali. Marathi dialect detector: Text and speech normalization to standard marathi and hindi. volume 5, pages 116–121, 2026. URL <http://ijisem.com>.
- [37] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. Self-instruct: Aligning language models with self-generated instructions, 2022.
- [38] Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B Hashimoto. Alpaca: A strong, replicable instruction-following model. *Stanford Center for Research on Foundation Models*, 2023.
- [39] Subhabrata Mukherjee, Arindam Mitra, Ganesh Jawahar, Sahaj Agarwal, Hamid Palangi, and Ahmed Awadallah. Orca: Progressive learning from complex explanation traces of gpt-4, 2023.
- [40] Cheng-Yu Hsieh et al. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. *arXiv preprint arXiv:2305.02301*, 2023.
- [41] Branislav Pecher et al. Better as generators than classifiers: Leveraging llms and synthetic data for low-resource multilingual classification. In *Findings of the Association for Computational Linguistics: EACL 2024*, 2024.
- [42] Ona de Gibert et al. Scaling low-resource mt via synthetic data generation with llms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024.
- [43] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):1–38, 2023.
- [44] Matt Post. A call for clarity in reporting bleu scores. In *Proceedings of WMT 2018*, pages 186–191, 2018. URL <https://doi.org/10.18653/v1/W18-6319>.